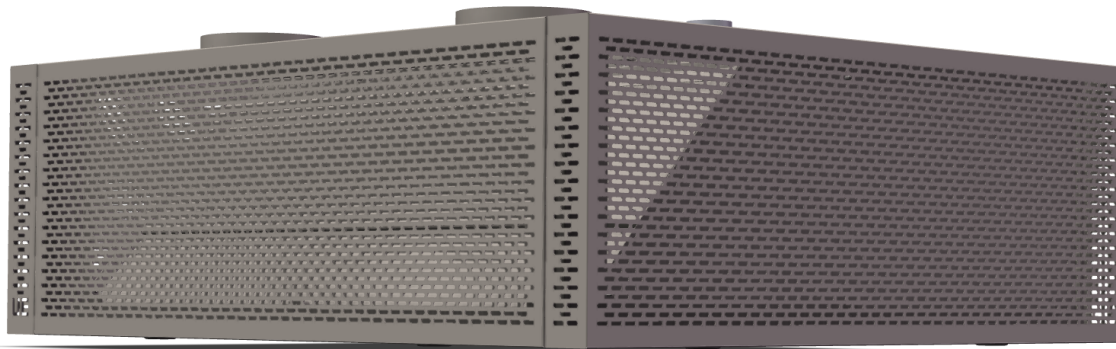




# OPTI COMMERCIAL KITCHEN VENTILATOR PROJECT SPECIFICATION

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Zachary Winter

Urban Environmental

25 Ra Ora Drive

Highbrook, Auckland

## OVERVIEW

Creating the highest quality commercial hospitality ventilation system requires in-depth knowledge of a chef's needs, restaurant servicing and expertise in ventilation engineering, HVAC install and modern manufacturing methods.

### Goals

1. Maximum comfort and efficiency for staff operations
2. The best air quality both in the kitchen and dining areas
3. Hygiene
4. Fire safety
5. Highest exhaust air quality
6. Energy savings

### Specification

All hoods shall be Urban Environmental OPTI models. Equipped with all standard OPTI features including make up air system to front and all exposed sides.

Optional and additional features will be added on a case by case basis to best meet the project goals to the highest standard.

Units shall be supplied boxed and complete, ready to be installed with all embedded technologies fully pre-wired from the factory.

#### Hood outer casing

- Constructed from 1.2 mm AISI 304 stainless steel in a brushed satin finish. The joints of the lower edge shall be fully welded. All exposed welds are ground and polished to the metal's original finish.
- Hood sides shall be of double-wall and removable to enable the air supply technology and cleaning.
- The hood shall be equipped with the **perimeter air curtain** and **adjustable front edge air knife** technology
- The **low velocity make up air** plenum modules will include two sets of **personal comfort nozzles** on the lower part of both the front and sides of the hoods where a wall is not present.
- The hood shall have a double skinned top to provide a heat barrier to the area above the hood in the event

## Exhaust and supply airflow rates

- The exhaust airflow rates shall be determined with a EN(1) 16282-1 based calculation method taking into account both the convective loads released by the cooking appliances and the hood capture efficiency according to ASTM 1704-12 standard. Both the exhaust airflow rates and capture efficiency shall be justified by a calculation note.
- Any modification of the hood installation height together with the input power, type and dimensions of the cooking appliances shall be brought to the attention of the hood manufacturer as they all significantly impact the exhaust airflow rates.
- The makeup air design, especially the balance between exhaust and supply, shall be entrusted to the hood manufacturer as it also impacts the exhaust airflow rates and capture efficiency. It is also key to preventing cross-contamination between the kitchen areas.

## Exhaust plenum and filters

- The exhaust plenum shall be constructed from 1.2 mm AISI 304 stainless steel in a brushed satin finish. The sides of the lower part shall be fully welded to be liquid-tight.
- It shall be equipped with **Triple Venturi Baffle Filter TVB(tm)** and **Micro Cyclonic Filter MCB(tm)** filters. Constructed from stainless steel, its efficiency shall be at least 95% on 8 microns particles or larger, as tested by an independent laboratory. The filters shall include UL classification and testing for non combustibility AS 1530.
- The exhaust connections shall be supplied with sliding balancing dampers which can be adjusted accurately from behind the light cover without otherwise modifying the airstream. The exhaust plenum shall be equipped with pressure taps for quick airflow measurement.

## Integrated makeup air and supply plenum

- For better staff comfort and optimised capture and containment efficiency of the hoods makeup air shall be introduced into the space at a very low velocity (less than 1.5 m/s).
- The hood shall be equipped with a rounded slot type perforated stainless steel front panel for optimum turbulence free air dispersion. This is combined with a second internal diffuser and calibrated structure to provide air distribution in the correct areas. This component shall be easy to remove for cleaning and maintenance operations.
- The supply connections shall be supplied with balancing dampers which can be adjusted accurately from behind the light cover without otherwise modifying the airstream.

## Light fitting

- Each hood shall be provided with a light fitting equipped with four LED downlights per module, providing at least 500 lux at the cooking appliances working level. It is protected with a stainless steel hatch and primary safety glass diffuser (the heat tolerance of the glass shall be -40 to 300°C). The fitting is IP65 with a tempered initial light diffuser and aluminium heat sink to rear.
- The hatch shall be removable for easy access to the LED fittings. It shall be held closed with recessed allen key type bolts.
- The light fitting shall provide **three different colour temperatures 2700K 4500K and 6000K** selectable on the light fittings.
- The LEDs life time shall be 50,000 hours. The power supply shall have at least the same life time and be DALI compatible.
- Lighting shall provide quality culinary light with a CRI Color Rendering Index of at least 80.

## Options

Engineers and designers may add requirements on a case by case basis to suit the specific operation and considering outdoor environments. Systems and testing for measured reductions and specific filtered equivalent exhaust levels in accordance to regulation are provided. Various technologies are available to achieve enhanced performance for certain advanced situations

- Ultraviolet
- Ozone injection
- Electrostatic filtration
- Heat recovery
- Cold misting
- Water washing

## Benefits

Operators can construct a hood of almost any size to suit any kitchen application. Modules bolt together end to end and back to back and in full ceiling ventilated ceiling configurations. Units can include custom extensions to get an exact mm fit to your kitchen.

- 30 to 40% reduction in exhaust airflow rates thanks to low velocity make up air and air curtain on all open sides and adjustable front air knife technology.
- High-efficiency TVB Triple Venturi Baffle filters (UL).
- High-efficiency MCB Micro Cyclonic Baffle compress filters (AS 1503).
- Prevention of the build-up of grease deposits, which pose a serious hygiene and fire hazard.
- Draught-free integrated supply air system to all open sides including adjustable air knife, air curtain and personal comfort nozzles for enhanced smoke capture and chef comfort
- Performance tested independently in accordance with the ASTM 1704 standard.
- Exhaust airflow rates based on VDI 2052 and a calculation method of the real heat loads.
- Effective commissioning due to the OPTI accurate adjustable damper system.
- Colour options allow architectural front of house installations to match surroundings

## Milestones

- Years of safe reliable operation
- Vibrant efficient working environment
- Significantly reduced energy and servicing costs
- Quality indoor air and external environmental preservation